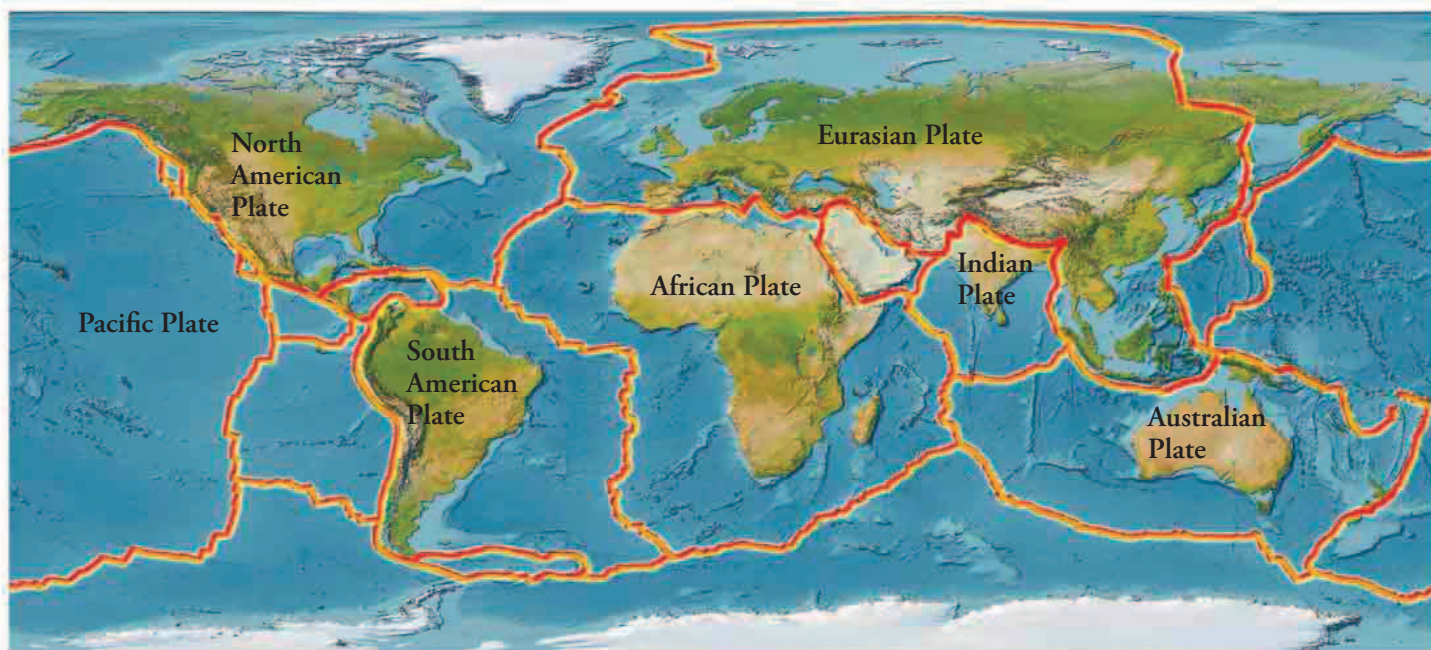
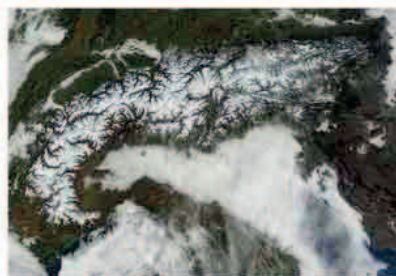




FRACTURED SHELL

The relentless movement of the mantle has broken the brittle lithosphere into sections that resemble fragments of a giant eggshell. The pieces are known as tectonic plates. Some are huge (the largest are labeled on the map above), but there are also many much smaller ones. Some plates form the ocean floors, while others underlie both continents and oceans.



▲ COLLIDING CRUST

The way the crust crumples up where plates push together is clear in this satellite view of the Alps.

PUSH AND PULL

When the convection currents rise through the mantle they push the plates apart, and when they sink back down again they pull the plates together. This means that the plates are always moving very slowly, at about the same rate as your fingernails grow. New crust is created where the plates pull apart, and old crust is crumpled up or destroyed where they push together.

CONTINENTAL DRIFT

The plates underlying the continents are constantly shifting relative to each other. Over vast spans of time, they are split up and pushed together in different arrangements. This is called continental drift. The continents that make up our present world are fragments of a giant supercontinent, Pangaea, that existed about 250 million years ago.

Pangaea



FRONTIER ZONES

The boundaries between colliding tectonic plates are the world's most active earthquake zones. They are also lined with chains of volcanoes. Some are more unstable than others—the boundary marked by the Indonesian islands of Sumatra and Java has more than 100 active volcanoes.

◀ FIRE ISLAND

Volcanoes cluster on Java, which has 45 active volcanoes and many dormant ones.



Rising mantle current